Name: Student ID:

Draw one number at random from the following set, and let $X$ be the value of the number that you draw.

$\{-2, -1, 0, 2\}$

1. Write down the pmf of $X$.

2. Draw the graph of the cdf of $X$.

Suppose $X \sim \text{Poisson}(6)$, where $X$ represents the number of students who earn an A+ in Stat 20, and $Y \sim \text{Poisson}(5)$ represents the number of students who earn an F in stat 20. Since our class is not curved, we can assume that these two quantities are independent.

3. What is the probability that *no one* fails Stat 20? Write an expression for this probability.

4. What is the probability that *at least* 10 students earn an A+ in Stat 20? Write an expression for this probability.

An American roulette wheel has 18 red, 18 black, and 2 green slots. On each spin, a white ball lands in one of the slots. Suppose you spin the wheel 50 times, and bet on red each time. Let $X$ be the number of times you win in 50 plays.

5. What is name of the probability distribution of $X$, with its parameters?

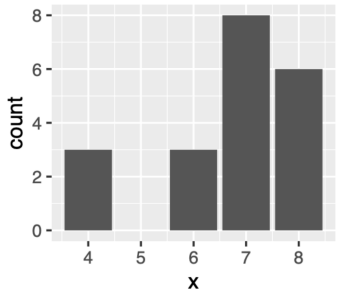
6. Write an expression for the probability that you win at least 12 times in the 50 spins.

7. Let $Y$ be a random variable corresponding to the result (win or lose) of the twenty-fifth game. What is the probability distribution of $Y$, with its parameter(s)?

Consider a deck of 52 cards. Suppose we are interested in the random variable $Z$, which corresponds to the number of hearts dealt in a hand of seven.

8. What is name of the probability distribution of $Z$, with its parameters?

9. The following is a visualization which displays twenty draws from a mysterious random variable $X$.




- Name a plausible probability distribution based on the draws. Choose from the following three options: *Bernoulli*, *Binomial*, *Poisson*.

- Give a plausible value(s) of the parameter(s) of the probability distribution you chose above.